📖 Detailed Paper Summaries

Below, each paper is displayed as a formatted card, making the title, authors, description, and potential applications easy to read.

In [ ]:

Modeling Long-term User Behaviors with Diffusion-driven Multi-interest Network for CTR Prediction

Authors: Weijiang Lai, Beihong Jin, Yapeng Zhang, Yiyuan Zheng, Rui Zhao, Jian Dong, Jun Lei, Xingxing Wang

Source File: `3705328.3748045.pdf`

> **Description**: This paper addresses the challenges of modeling long-term user behaviors for Click-Through Rate (CTR) prediction, specifically the issues of vast behavior numbers and noise interference. The authors propose DiffuMIN (Diffusion-driven Multi-Interest Network), a two-stage model that first extracts diverse user interests using a target-oriented method and then uses a diffusion module to generate augmented interests to explore the user's latent interest space. The key contributions include a target-oriented multi-interest extraction method and a novel diffusion module guided by contextual interests, which was shown to improve CTR and CPM in online A/B testing.

Potential Applications:

- Improving Click-Through Rate (CTR) prediction in recommender systems and online advertising
- Personalized ad ranking on social media platforms to increase user clicks
- Dynamic product recommendations in e-commerce based on extensive user browsing history

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Lasso: Large Language Model-based User Simulator for Cross-Domain Recommendation

Authors: Yue Chen, Susen Yang, Tong Zhang, Chao Wang, Mingyue Cheng, Chenyi Lei, Han Li

Source File: `3705328.3748048.pdf`

> **Description**: This paper addresses the data efficiency problem in Cross-Domain Recommendation (CDR), where existing methods require a substantial number of overlapping users to mitigate the cold-start issue. The authors propose Lasso, a framework that uses a Large Language Model (LLM) as a user simulator to generate interactions for cold-start users in a target domain based on their source domain history. To improve efficiency and accuracy, Lasso incorporates a Personalized Candidate Pool (PCP) built with collaborative filtering and a Confidence-Guided Inference (CGI) module to filter noisy simulated data.

Potential Applications:

- E-commerce platforms like Amazon to recommend products to new users based on their activities in a different product category (e.g., book preferences to recommend movies)
- Movie recommendations on platforms like Netflix for new users by leveraging their book reading history from another service
- Music recommendations on Spotify for users with sparse listening data by simulating preferences based on their activity on a concert ticketing platform

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PinFM: Foundation Model for User Activity Sequences at a Billion-scale Visual Discovery Platform

Authors: Xiangyi Chen, Kousik Rajesh, Matthew Lawhon, Zelun Wang, Hanyu Li, Haomiao Li, Saurabh Vishwas Joshi, Pong Eksombatchai, Jaewon Yang, Yi-Ping Hsu, Jiajing Xu, Charles Rosenberg

Source File: `3705328.3748050.pdf`

> **Description**: This paper introduces PinFM, a foundational model for understanding user activity sequences across multiple applications on a large-scale visual discovery platform. The model addresses challenges such as scalability, latency, cost constraints, and handling new items by pre-training a large transformer on extensive user activity data and then fine-tuning it for specific applications. Key contributions include novel infrastructure and algorithmic optimizations like the Deduplicated Cross-Attention Transformer (DCAT) to improve throughput, and techniques to capture interactions between user sequences and candidate items, which led to a 20% increase in engagement with new items.

Potential Applications:

- Improving Home Feed recommendations on a visual discovery platform like Pinterest
- Enhancing Related Items Feed recommendations on a visual discovery platform like Pinterest
- Personalized content feeds on social media platforms like Instagram, using a user's entire interaction history
- Product recommendations on e-commerce sites, where the model can be fine-tuned for different sections like 'daily deals' or 'new arrivals'

Privacy-Preserving Social Recommendation: Privacy Leakage and Countermeasure

Authors: Yuyue Chen, Peng Yang, Zoe Lin Jiang, Wenhao Wu, Junbin Fang, Xuan Wang, Chuanyi Liu

Source File: `3705328.3748051.pdf`

> **Description**: This paper identifies a critical privacy vulnerability in the Secure Social Recommendation (SeSoRec) framework, which allows a rating platform and a social platform to collaborate without sharing raw data. The authors demonstrate that the framework's core protocol (SSMM) leaks intermediate information, enabling the reconstruction of the private social graph due to the reuse of a one-time pad key. The paper proposes two attacks, the Two-Time Pad Attack and the Zero-Padding Attack, which can successfully reconstruct a high percentage of the private data, and introduces a secure alternative, the Privacy-Preserving Matrix Multiplication (PPMM) protocol, to eliminate the leakage.

Potential Applications:

- Secure social recommendation systems where an e-commerce platform (e.g., Amazon) uses social graph data from another platform (e.g., Facebook) without direct data sharing
- Privacy-preserving personalized advertising where user data from different services are combined without revealing private inputs
- Secure multi-party computation for financial institutions to build joint risk models based on separate customer datasets

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A Language Model-Based Playlist Generation Recommender System

Authors: Enzo Charolois-Pasqua, Eléa Vellard, Youssra Rebboud, Pasquale Lisena, Raphaël Troncy

Source File: `3705328.3748053.pdf`

> **Description**: This paper presents a novel approach for generating music playlists using only a playlist title as input, addressing the challenge of creating themed playlists in cold-start scenarios. The proposed pipeline leverages language models by first clustering existing playlists based on the semantic content of their tracks and then fine-tuning a transformer model on these thematic clusters. Recommendations are generated by finding the most similar existing playlists to a new title and using a voting mechanism to select the most frequent tracks.

Potential Applications:

- Automatic playlist generation on music streaming services like Spotify based on a user-provided title (e.g., 'Spring awakening' or 'Country summer')
- Cold-start music recommendations for new users who have not yet provided any listening history
- Enhancing collaborative filtering systems by adding a semantic constraint based on the playlist title to improve song suggestions

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You Don’t Bring Me Flowers: Mitigating Unwanted Recommendations Through Conformal Risk Control

Authors: Giovanni De Toni, Erasmo Purificato, Emilia Gomez, Andrea Passerini, Bruno Lepri, Cristian Consonni

Source File: `3705328.3748054.pdf`

> **Description**: This paper addresses the issue of recommender systems propagating unwanted or harmful content, which degrades user satisfaction and trust. The authors propose a model-agnostic and distribution-free method using conformal risk control to provably limit the amount of unwanted content in recommendations by leveraging simple binary user feedback. The method filters potentially unsafe items and replaces them with previously consumed, safe content, ensuring a user-defined risk level is met while maintaining the size of the recommendation set.

Potential Applications:

- Filtering unwanted or harmful videos on video-sharing platforms like YouTube or Kuaishou based on user reports
- Providing users with transparent and controllable content filters on social media feeds
- Improving content moderation in e-commerce to prevent the recommendation of irrelevant or low-quality products

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MoRE: A Mixture of Reflectors Framework for Large Language Model-Based Sequential Recommendation

Authors: Weicong Qin, Yi Xu, Weijie Yu, Chenglei Shen, Xiao Zhang, Ming He, Jianping Fan, Jun Xu

Source File: `3705328.3748055.pdf`

> **Description**: This paper addresses limitations in current LLM-based sequential recommendation methods, such as the failure to decouple explicit features from implicit patterns, underutilization of collaborative filtering signals, and inefficient update strategies. The authors propose MoRE (Mixture of REflectors), a framework that uses three perspective-aware offline reflection processes to disentangle intra-user (explicit/implicit) and cross-user (collaborative) patterns. MoRE's meta-reflector employs a self-improving strategy and a contextual bandit mechanism to dynamically select the optimal reflection for each user, improving recommendation performance with minimal overhead.

Potential Applications:

- Personalized product recommendations on e-commerce sites that distinguish between a user's interest in a product category (explicit) and brand loyalty (implicit)
- Movie or music track recommendations that incorporate collaborative filtering signals by analyzing similar users' preferences
- Dynamic content recommendations on news platforms that adapt to evolving user interests over time

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NLGCL: Naturally Existing Neighbor Layers Graph Contrastive Learning for Recommendation

Authors: Jinfeng Xu, Zheyu Chen, Shuo Yang, Jinze Li, Hewei Wang, Wei Wang, Xiping Hu, Edith Ngai

Source File: `3705328.3748059.pdf`

> **Description**: This paper addresses the limitations of existing Graph Contrastive Learning (GCL) methods in recommender systems, which often rely on data augmentation techniques that introduce noise and increase computational costs. The authors propose NLGCL, a novel contrastive learning framework that leverages the naturally existing contrastive views between neighbor layers within Graph Neural Networks (GNNs). By treating each node and its neighbors in the next layer as positive pairs, NLGCL avoids augmentation, preserving semantic relevance and improving both effectiveness and efficiency.

Potential Applications:

- Improving collaborative filtering in e-commerce for more accurate product suggestions by leveraging high-order user-item relationships
- Enhancing social media content recommendations by better capturing user preferences from sparse interaction data
- Optimizing short video recommendations on platforms like TikTok by efficiently modeling user-item interactions without costly data augmentation

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Test-Time Alignment with State Space Model for Tracking User Interest Shifts in Sequential Recommendation

Authors: Changshuo Zhang, Xiao Zhang, Teng Shi, Jun Xu, Ji-Rong Wen

Source File: `3705328.3748060.pdf`

> **Description**: This paper addresses the challenge of user interest shifts in sequential recommendation, where models trained on static data often fail to adapt to dynamic user behavior during testing. The authors propose T2ARec, a novel model that leverages Test-Time Training (TTT) and a State Space Model (SSM) backbone to track these shifts. T2ARec introduces two test-time alignment modules: one for aligning time intervals to capture temporal dynamics, and another for aligning interest states to explicitly identify preference shifts, allowing for real-time model parameter updates.

Potential Applications:

- Adaptive e-commerce recommendations that adjust to a user's changing interests (e.g., from work-related items on weekdays to leisure items on weekends)
- Personalized content feeds on streaming platforms that update in real-time based on a user's immediate viewing behavior
- Dynamic news recommendations that adapt to breaking events and evolving user interests

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A Multi-Factor Collaborative Prediction for Review-based Recommendation

Authors: Junrui Liu, Tong Li, Mingliang Yu, Shiqiu Yang, Zifang Tang, Zhen Yang

Source File: `3705328.3748062.pdf`

> **Description**: This paper addresses the underutilized potential of rating behaviors for improving Click-Through Rate (CTR) prediction in review-based recommender systems. The authors propose a multi-factor collaborative prediction method (MFC) that models the complex relationship between click and rating behaviors by factorizing it into linear, sharing, and cross-correlation components. MFC extracts click, rating, and shared factors from user reviews and combines them to make accurate CTR predictions, using a novel regularization method to ensure the rating factors are learned accurately.

Potential Applications:

- Improving Click-Through Rate (CTR) prediction for products on online shopping platforms by analyzing user ratings and reviews
- Enhancing ad-targeting systems by predicting click probability based on both implicit user clicks and explicit product ratings
- Optimizing movie recommendations by using textual reviews and star ratings to predict which trailers users are most likely to watch

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Recommendation and Temptation

Authors: Md Sanzeed Anwar, Paramveer S. Dhillon, Grant Schoenebeck

Source File: `3705328.3748063.pdf`

> **Description**: This paper challenges traditional recommender systems by proposing a novel design that models the tension between long-term value (enrichment) and short-term appeal (temptation) in user choices. The authors introduce a behavioral model that accounts for how both factors, along with off-platform alternatives, influence consumption decisions. Based on this model, they formulate an objective to maximize consumed enrichment and propose a locally greedy recommendation strategy, supported by an estimation framework that uses both explicit feedback and implicit choice data.

Potential Applications:

- Video recommendations on platforms like YouTube or Netflix that balance engaging content with more enriching or educational options to improve long-term user satisfaction
- News feed curation that prioritizes in-depth, valuable articles over clickbait to foster a more informed user base
- Educational app recommendations (e.g., Duolingo) that make learning engaging enough to compete with more tempting social media apps

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VL-CLIP: Enhancing Multimodal Recommendations via Visual Grounding and LLM-Augmented CLIP Embeddings

Authors: Ramin Giahi, Kehui Yao, Sriram Kollipara, Kai Zhao, Vahid Mirjalili, Jianpeng Xu, Topojoy Biswas, Evren Korpeoglu, Kannan Achan

Source File: `3705328.3748064.pdf`

> **Description**: This paper addresses key limitations of vision-language models like CLIP in e-commerce recommendation, including weak object-level alignment, ambiguous text representations, and domain mismatch. The authors propose VL-CLIP, a framework that enhances CLIP embeddings by integrating Visual Grounding to localize key product attributes in images and using an LLM-based agent to generate enriched, structured text descriptions. This approach was shown to significantly improve retrieval accuracy and recommendation quality on a large e-commerce platform, increasing CTR, ATC, and GMV.

Potential Applications:

- Enhanced visual search on e-commerce platforms like Walmart.com for complex queries such as 'area rug with pet pic'
- Visually coherent product recommendations in fashion and home goods categories
- Improved multimodal retrieval for large-scale product catalogs with tens of millions of items

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How Do Users Perceive Recommender Systems' Objectives?

Authors: Patrik Dokoupil, Ludovico Boratto, Ladislav Peska

Source File: `3705328.3748066.pdf`

> **Description**: This paper investigates the mismatch between how multi-objective recommender systems (MORS) measure objectives like relevance, novelty, and diversity, and how users actually perceive them. Through a large user study in the book and movie domains, the authors find that while established metrics correlate with user perceptions to some extent, the mapping is imperfect and there are conceptual differences in how users understand these objectives. The study suggests that aligning the system's interpretation with the user's conceptual understanding can significantly reduce this mismatch, highlighting a need for better user education and domain-specific metrics.

Potential Applications:

- Designing user-controllable interfaces for movie and book recommender systems that allow users to tune objectives like novelty and diversity
- Improving user satisfaction and trust in personalized recommendation platforms by better aligning algorithmic goals with user perceptions
- Developing more accurate offline evaluation metrics for recommender systems that better reflect real-world user experience

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R4ec: A Reasoning, Reflection, and Refinement Framework for Recommendation Systems

Authors: Hao Gu, Rui Zhong, Yu Xia, Wei Yang, Chi Lu, Peng Jiang, Kun Gai

Source File: `3705328.3748068.pdf`

> **Description**: This paper addresses the limitation of existing LLM-based recommender systems, which primarily rely on intuitive 'System-1' thinking and are sensitive to reasoning errors. The authors propose R4ec, a framework that introduces 'System-2' thinking through an iterative reasoning, reflection, and refinement process. The framework uses two models: an 'actor' model for reasoning and a 'reflection' model to judge the actor's responses and provide feedback, enabling the actor to refine its knowledge and improve recommendation quality.

Potential Applications:

- Enhancing movie and book recommendations by incorporating refined user preferences and factual item knowledge into a recommendation backbone
- Improving performance in online advertising platforms, as demonstrated by a 2.2% increase in revenue in an A/B test
- More reliable knowledge extraction for personalized user preference modeling in any domain requiring deep reasoning

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Enhancing Online Video Recommendation via a Coarse-to-fine Dynamic Uplift Modeling Framework

Authors: Chang Meng, Chenhao Zhai, Xueliang Wang, Shuchang Liu, Xiaoqiang Feng, Lantao Hu, Xiu Li, Han Li, Kun Gai

Source File: `3705328.3748070.pdf`

> **Description**: This paper tackles challenges in video recommendation, specifically managing content distribution and modeling dynamic user preferences, by applying uplift modeling. The authors propose a Coarse-to-fine Dynamic Uplift Modeling (CDUM) framework, which consists of a coarse-grained module for modeling long-term preferences using offline features and a fine-grained module for capturing real-time interests with online features. This two-module approach dynamically identifies user groups and applies treatments (adjustments to video type distribution) to enhance user experience and retention.

Potential Applications:

- Managing the distribution of short and long videos in recommendation feeds on platforms like Kuaishou to improve user experience and retention
- Personalized content curation in streaming services to ensure a diverse mix of content types based on both long-term habits and real-time user behavior
- Optimizing online advertising campaigns by dynamically targeting user groups based on their predicted response to different ad formats

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GenSAR: Unifying Balanced Search and Recommendation with Generative Retrieval

Authors: Teng Shi, Jun Xu, Xiao Zhang, Xiaoxue Zang, Kai Zheng, Yang Song, Enyun Yu

Source File: `3705328.3748071.pdf`

> **Description**: This paper addresses the performance trade-off between search and recommendation (S&R) tasks in joint modeling, which arises from their different information needs (semantic relevance for search, collaborative relationships for recommendation). The authors propose GenSAR, a method that unifies S&R using generative retrieval, which represents items as identifiers. GenSAR designs joint S&R identifiers to incorporate both semantic and collaborative information and uses varied prompts in a unified sequence-to-sequence framework to help the model understand the distinct requirements of each task, thereby mitigating the trade-off.

Potential Applications:

- Jointly optimizing search and recommendation on e-commerce platforms to provide more relevant results for both queried searches and personalized suggestions
- Improving content discovery on video or music platforms by balancing semantic query matching with collaborative filtering based on user history
- Creating a unified user model for commercial apps that offer both search and recommendation functionalities, leading to a more coherent user experience

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Hierarchical Graph Information Bottleneck for Multi-Behavior Recommendation

Authors: Hengyu Zhang, Chunxu Shen, Xiangguo Sun, Jie Tan, Yanchao Tan, Yu Rong, Hong Cheng, Lingling Yi

Source File: `3705328.3748073.pdf`

> **Description**: This paper addresses two key challenges in multi-behavior recommendation: severe distribution disparities across behaviors and negative transfer effects from noisy auxiliary behaviors. The authors propose a model-agnostic Hierarchical Graph Information Bottleneck (HGIB) framework that learns compact yet sufficient representations, preserving essential information for target behavior prediction while filtering redundancies. To further reduce noise, the framework includes a Graph Refinement Encoder (GRE) that dynamically prunes redundant edges from the interaction graph.

Potential Applications:

- E-commerce recommendations that use auxiliary behaviors like 'view' and 'add-to-cart' to improve prediction of the 'buy' behavior
- News recommendation on social platforms like WeChat, using clicks and likes to better predict shares
- Live-streaming recommendation that leverages user interactions such as clicks, likes, and comments to predict which streams will receive gifts

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LLM-RecG: A Semantic Bias-Aware Framework for Zero-Shot Sequential Recommendation

Authors: Yunzhe Li, Junting Wang, Hari Sundaram, Zhining Liu

Source File: `3705328.3748077.pdf`

> **Description**: This paper addresses the challenge of domain semantic bias in LLM-based Zero-shot Cross-domain Sequential Recommendation (ZCDSR), where differences in vocabulary and content focus between domains hinder knowledge transfer. The authors propose LLM-RecG, a framework that improves cross-domain alignment at both item and sequential levels. At the item level, it uses a generalization loss to balance inter-domain compactness and intra-domain diversity, while at the sequential level, it transfers behavioral patterns by clustering source domain sequences and using attention-based aggregation for target domain inference.

Potential Applications:

- E-commerce recommendations for new product categories by transferring user preferences from established categories (e.g., from 'Industrial & Scientific' to 'Video Games')
- Personalized content recommendations for users entering a new domain on a streaming platform without any interaction history in that domain
- Adapting a recommendation model from one platform to another with different content, such as from Amazon to Steam

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Mapping Stakeholder Needs to Multi-Sided Fairness in Candidate Recommendation for Algorithmic Hiring

Authors: Mesut Kaya, Toine Bogers

Source File: `3705328.3748079.pdf`

> **Description**: This paper addresses the lack of a multi-stakeholder perspective in fairness for algorithmic hiring, where past analyses have focused solely on the job seeker. Through semi-structured interviews with 40 stakeholders—including job seekers, recruiters, companies, and recruitment agency employees—the authors explore lived experiences of unfairness and co-design fairness definitions. The study maps these diverse and sometimes conflicting needs to existing fairness metrics, providing a framework for multi-sided fairness in a candidate recommender system for a human-in-the-loop hiring scenario.

Potential Applications:

- Designing fair candidate recommender systems for recruitment agencies to help recruiters identify suitable candidates for job postings
- Auditing algorithmic hiring tools to ensure they meet fairness criteria from the perspective of job seekers, employers, and recruiters
- Developing fairness-aware job recommendation platforms that balance the needs of different user groups (e.g., ensuring diverse candidates are shown to companies)

USB-Rec: An Effective Framework for Improving Conversational Recommendation Capability of Large Language Model

Authors: Jianyu Wen, Jingyun Wang, Cilin Yan, Jiayin Cai, Xiaolong Jiang, Ying Zhang

Source File: `3705328.3748080.pdf`

> **Description**: This paper proposes a training-inference framework, USB-Rec, to improve the intrinsic conversational recommendation capabilities of Large Language Models (LLMs) at the model level. The framework introduces a Preference Optimization dataset construction strategy that uses an LLM-based user simulator to score and select high-quality responses for Reinforcement Learning (RL) training. Additionally, a Self-Enhancement Strategy (SES) is used during inference to further exploit the model's learned potential by sampling multiple responses and selecting the best one through an internal scoring mechanism.

Potential Applications:

- Improving conversational recommender systems for movie recommendations, as demonstrated on the ReDial dataset
- Enhancing multi-domain conversational recommenders for books and movies, as shown with the OpenDialkg dataset
- Developing more streamlined and efficient chatbot-based shopping assistants that can be fine-tuned to understand user preferences better

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IP2: Entity-Guided Interest Probing for Personalized News Recommendation

Authors: Youlin Wu, Yuanyuan Sun, Xiaokun Zhang, Haoxi Zhan, Bo Xu, Liang Yang, Hongfei Lin

Source File: `3705328.3748091.pdf`

> **Description**: This paper addresses the oversight of entities' unique role in news recommendation, aligning with behavioral science findings that news reading involves scanning, title reading, and clicking. The authors propose IP2, a method to probe entity-guided interest at both intra- and inter-news levels. At the intra-news level, a Transformer-based encoder and contrastive pre-training are used to identify signature entities within a title, while at the inter-news level, a dual-tower user encoder with cross-attention captures interest from both semantic and entity perspectives.

Potential Applications:

- Personalized news recommendation on platforms like MSN News, as demonstrated with the MIND dataset
- Improving news feed curation by better understanding user interest based on key entities (people, places) mentioned in headlines
- Content recommendation for online media platforms that can better predict user clicks by modeling the interaction between semantic content and highlighted entities

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Breaking Knowledge Boundaries: Cognitive Distillation-enhanced Cross-Behavior Course Recommendation Model

Authors: Ruoyu Li, Yangtao Zhou, Chenzhang Li, Hua Chu, Jianan Li, Yuhan Bian

Source File: `3705328.3748083.pdf`

> **Description**: This paper addresses challenges in integrating Cognitive Diagnosis (CD) and Course Recommendation (CR) on online education platforms, such as divergent task objectives and inconsistent data magnitudes. The proposed C3Rec model transfers knowledge of learners' cognitive states from learner-exercise interactions to enhance course recommendations. C3Rec uses a dual-granularity cognitive diagnosis module for accurate assessment, a cross-behavior module for semantic alignment, and a triple-stage distillation mechanism to use cognitive states as prior knowledge for profiling course preferences.

Potential Applications:

- Personalized course recommendations on MOOC platforms like XuetangX
- Adaptive learning systems that suggest courses based on a learner's mastery of specific knowledge concepts, as identified through their exercise performance
- Enhancing learning efficiency on intelligent education platforms like Santa by providing tailored learning paths

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Affect-aware Cross-Domain Recommendation for Art Therapy via Music Preference Elicitation

Authors: Bereket A. Yilma, Luis A. Leiva

Source File: `3705328.3748084.pdf`

> **Description**: This paper explores the use of Cross-Domain Recommendation (CDR) to enhance Art Therapy (AT) by leveraging music to elicit user preferences for visual art recommendations. The authors propose a family of CDR methods that map affective properties from music to paintings, addressing the limitation of current visual-only systems. A large-scale user study demonstrated that music-driven preference elicitation is effective for generating therapeutic artwork recommendations, performing competitively with traditional visual-only approaches.

Potential Applications:

- Personalizing therapeutic artwork recommendations for patients undergoing Art Therapy to facilitate emotional processing and recovery
- Digital mental health interventions that use music preferences to suggest other forms of therapeutic content, such as visual art or films
- Enhancing user experience in online art galleries by recommending paintings based on the mood or emotion of music a user is listening to

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Heterogeneous User Modeling for LLM-based Recommendation

Authors: Honghui Bao, Wenjie Wang, Xinyu Lin, Fengbin Zhu, Teng Sun, Fuli Feng, Tat-Seng Chua

Source File: `3705328.3748085.pdf`

> **Description**: This paper addresses challenges in modeling user preferences from heterogeneous behaviors across multiple domains for open-domain recommendation, such as poor generalization and the 'domain seesaw' phenomenon. The authors propose HUM (Heterogeneous User Modeling), a method that enhances LLM-based recommendation with a compression enhancer and a robustness enhancer. The compression enhancer uses a custom prompt and masking to compress noisy interactions into a tailored token, while the robustness enhancer uses a domain importance score to guide optimization and mitigate domain conflicts.

Potential Applications:

- Open-domain recommendation systems that model user preferences across diverse platforms like e-commerce, social media, and life service apps
- Multi-domain e-commerce recommendations (e.g., across 'Scientific', 'Office', 'Books' categories on Amazon) that avoid performance degradation in one domain when another is improved
- Personalized recommendations for users with sparse data by generalizing preferences learned from their activities in other, more data-rich domains

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Integrating Individual and Group Fairness for Recommender Systems through Social Choice

Authors: Amanda Aird, Elena Štefancová, Anas Buhayh, Cassidy All, Martin Homola, Nicholas Mattei, Robin Burke

Source File: `3705328.3748087.pdf`

> **Description**: This paper addresses the limitation of fairness-aware recommender systems that typically focus on a single fairness concern, by proposing a general solution using social choice methods to integrate multiple, heterogeneous fairness definitions. The authors extend a prior framework (SCRUF-D) to incorporate both provider-side individual and group fairness objectives, demonstrating that these can be optimized jointly. The study identifies both synergies and tensions between individual and group fairness objectives, showing their correlation depends on the specific groups and dataset characteristics.

Potential Applications:

- Developing fair recommender systems for music streaming that balance individual artist exposure (individual fairness) with representation of artists from specific demographic groups (group fairness)
- Fair movie recommendation on platforms like MovieLens that considers both individual content creators and groups (e.g., movies with female writers/directors or non-English films)
- Algorithmic hiring platforms to ensure fairness for both individual candidates and demographic groups

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Beyond Immediate Click: Engagement-Aware and MoE-Enhanced Transformers for Sequential Movie Recommendation

Authors: Haotian Jiang, Sibendu Paul, Haiyang Zhang, Caren Chen

Source File: `3705328.3748089.pdf`

> **Description**: This paper proposes a sequential recommendation framework to address several key issues in video streaming services, including suboptimal negative sampling, fixed context lengths, and single-task optimization. The framework introduces an adaptive Mixture of Experts (MoE) architecture to handle different time scales, a personalized hard negative sampling strategy based on completion rates, and a multi-task learning approach with an engagement-aware loss. It also extends next-item prediction to a next-K forecasting task, improving both immediate and multi-step predictions on large-scale streaming data.

Potential Applications:

- Sequential movie and video recommendations on streaming services like Amazon Prime Video
- Improving user engagement and retention on platforms like Netflix, Hulu, or YouTube by optimizing for completion rates alongside click-through rates
- Predicting a user's next sequence of viewed content (next-K items) for better session-based personalization

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In [ ]: